

- 5 (a) progressive: all particles have same amplitude
stationary: no nodes or antinodes or maximum to minimum/zero amplitude B1
- progressive: adjacent particles are not in phase
stationary: waves particles are in phase (between adjacent nodes) B1 [2]
- (b) (i) wavelength 1.2 m (zero displacement at 0.0, 0.60 m, 1.2 m, 1.8 m, 2.4 m)
- either peaks at 0.30 m and 1.5 m and troughs at 0.90 m and 2.1 m
or vice versa (but not both) B1
- maximum amplitude 5.0 mm B1 [2]
- (ii) 180° or π rad A1 [1]
- (iii) at $t = 0$ particle has kinetic energy as particle is moving B1
- at $t = 5.0$ ms no kinetic energy as particle is stationary
so decrease in kinetic energy (between $t = 0$ and $t = 5.0$ ms) B1 [2]